

An Overlay for Social Networks based on Directed Acyclic Graphs on the Edge

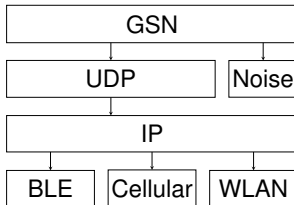
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Chair of Connected Mobility
School of Computation, Information, and Technology
Technical University of Munich

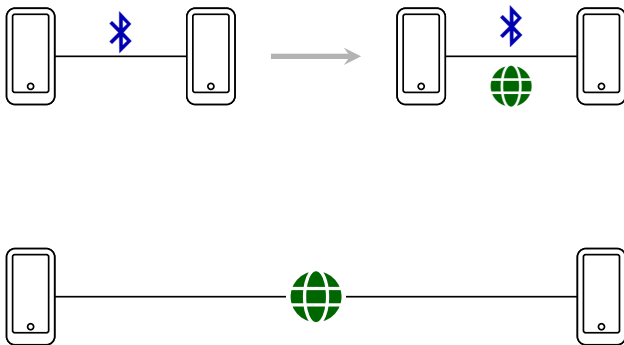


Architecture



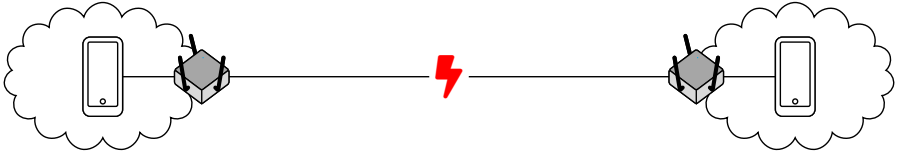
- Connect two peer devices according to priority and proximity pair
- Dual-stack network comms: BLE first, Cellular (5G/LTE) and WLAN second
- In close range, use BLE for signaling to establish UDP connection
- ICE-similar stack for address reflection, hole-punching using friends

Dual-stack comms

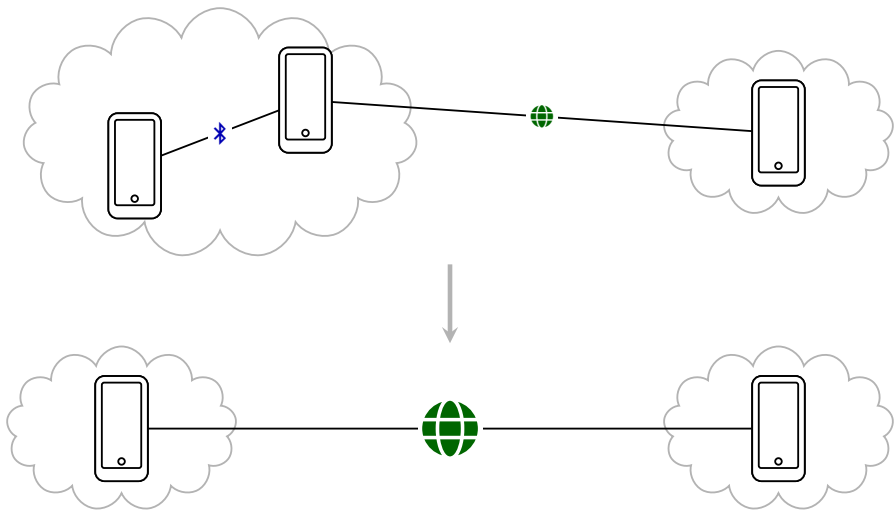


- BLE as signaling channel
- Advertise multiple addresses
- UDP for hole-punching
- UDX for long-range communication

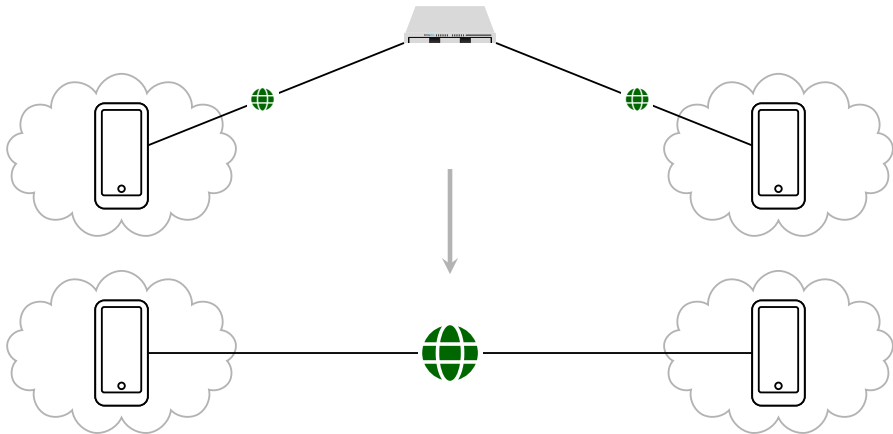
NAT



Well-connected friends



Rendezvous Server



Since mid-April: what we converged on

- Built the transport around cryptographic identity: Ed25519 public keys as peer identity and signed packets as the basic trust object.
- Stabilized the dual-stack direction: BLE discovery and nearby direct messaging, with UDP/UDX explored for longer-range connectivity.
- Worked through NAT traversal: public address discovery, hole-punching, well-connected friends, and rendezvous-style coordination.
- Added delivery machinery: ACK/read receipts, fragmentation for large payloads, Redux-backed peer/transport/message/friendship state.
- Added Noise sessions as hop-level channel security, then identified why relaying needs a separate end-to-end sealed message envelope.
- Reframed the thesis around a BLE-only path: remove UDP, analyze friend-based relaying, and compare it with direct BLE and open stranger relay.
- Connected the research plan to opportunistic networking, Sybil/misbehavior defenses, and Shapiro–Keidar work on grassroots/blocklace systems.

Mobility model: BLE-only relay

Goal

- Test if BLE relay can replace UDP locally.
- Compare direct, friend, and open relay (simulated).
- Report delivery, latency, overhead, and abuse surface.

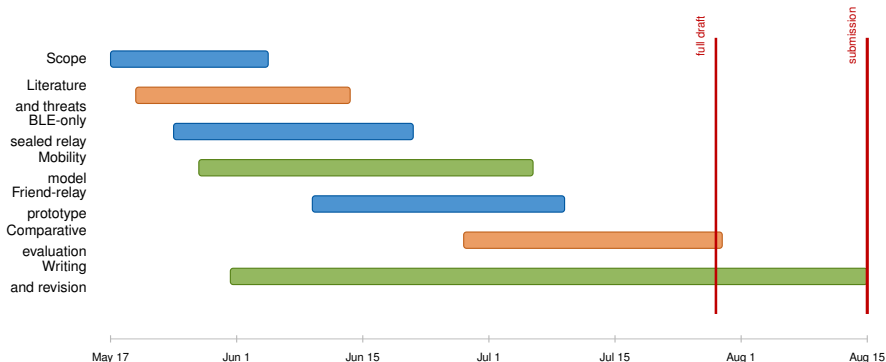
What we track

- Encounters: pseudonym pair, start/end, duration, inter-contact, RSSI band.
- Delivery: created, relayed, ACKed, hop count, TTL expiry, size, failure.
- Cost: relay bytes, duplicates, queues, scan/advertise time, battery proxy.

How

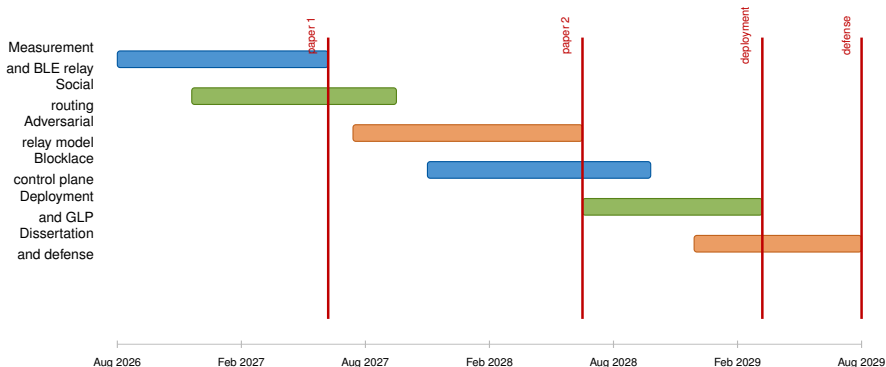
- Encrypted local trace log.
- Upload only by explicit opt-in.
- Replay traces in simulator calibrated by BLE contacts.
- No content; coarse location only by separate opt-in.

Master thesis gantt: submission on 15 Aug 2026



- Minimum contribution: secure relay design, mobility model, comparison, and critical security analysis.

PhD runway: three-year research arc



- Thesis-to-PhD bridge: from "does BLE relay work?" to "can grassroots social trust make mobile relaying secure and abuse-resistant?"